

DAVIS: A Depth-Only End-to-End Active-Vision Framework for Humanoid Soccer Skills

Anonymous Submission

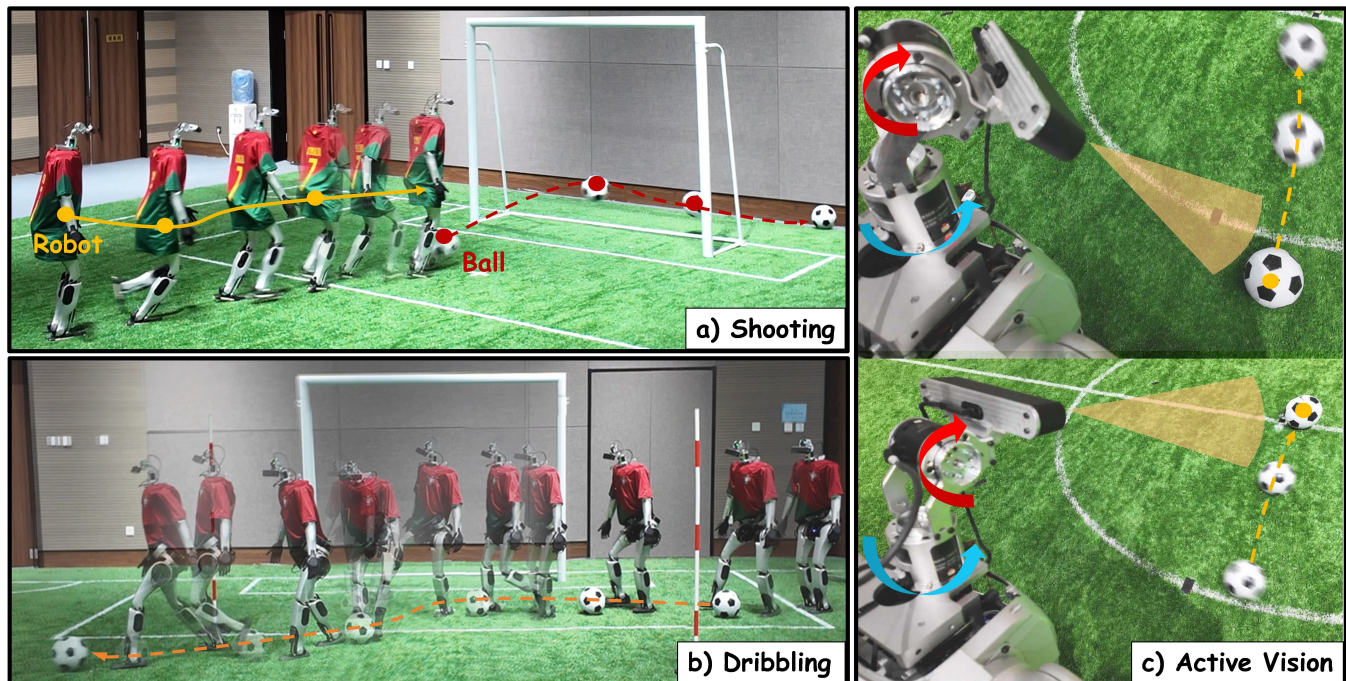


Fig. 1. Overview of our humanoid soccer system. (a) The robot performs goal-oriented shooting by approaching the ball and kicking it into the goal. (b) The robot demonstrates agile multi-directional dribbling with continuous ball control, including turning and weaving around poles. (c) Our active vision design enables the robot to actively reorient its head in yaw and pitch to keep the ball near the center of the field of view, thereby improving perception-action coordination.

Abstract—Humanoid soccer contact skills require more than producing high-impact foot-ball contacts: the robot must close the loop over perception, approach, alignment, impact, and recovery while its own motion induces substantial viewpoint changes, frequent loss of the ball from view, and uncertain contact outcomes. In this work, we ask a compact yet stricter question: can a humanoid learn soccer contact skills using only a head-mounted depth image, proprioceptive history, and an optional low-dimensional task command, and directly output 25-DoF joint PD targets without extra runtime perception or planning modules? To this end, we propose DAVIS, a depth-only end-to-end framework for humanoid soccer skills that learns visibility-aware auxiliary geometry during training, and combines GT-to-prediction annealing, task curricula, and AMP-style motion priors to smoothly bridge privileged supervision and real deployment. Built on this framework, we instantiate representative soccer contact skills, including goal-directed shooting and directional dribbling, through task-specific definitions of objects, commands, rewards, and curricula, and validate them through simulation, Noetix E1 real-robot experiments, and ablations. Additional online material, including result videos, is available on the anonymous project page at <https://anonymous.4open.science/w/anonymous-davis-review-B71B/>.

I. INTRODUCTION

Humanoid robots have made substantial progress in reinforcement-learning-driven dynamic locomotion [1]–[6],

but soccer remains a harder setting because it requires the robot to perceive, approach, align, contact, and recover while the target and the robot itself are moving. Unlike standard locomotion benchmarks, soccer contact skills require the policy to reason not only about *where the ball is*, but also about *how to approach it*, *how to keep the target in view*, and *how contact should change the subsequent ball motion*.

On real humanoids, onboard sensing and whole-body control further amplify these difficulties: the head-mounted view changes with body motion, active head yaw and pitch determine what evidence the policy receives, and small perception or stance errors can change the contact outcome [7]–[11]. Existing soccer systems often manage this complexity with modular perception, localization, planning, and skill-switching interfaces, including RGB detectors, LiDAR/RGB-D-based localization, or external state inputs. Because soccer contact skills depend on local ball and goal geometry, head-mounted depth provides a direct geometric observation from which to study control without externally supplied scene state. This motivates the question: **can a humanoid learn deployable soccer contact skills from only head-mounted depth, proprioceptive history, and low-dimensional task commands — without any external scene state?**

We propose **DAVIS**, a depth-only end-to-end reinforce-

ment learning framework to answer this question. At deployment, the policy remains a closed-loop mapping from head-mounted depth, proprioceptive history, and low-dimensional commands to 25-DoF joint PD targets. This is achieved by placing structure inside the learning signal rather than in the deployment pipeline, following the idea of using privileged simulation information to guide deployable policies [12]–[14]. Auxiliary geometry and visibility predictions are learned from privileged simulation supervision, while the actor gradually transitions from stable geometric guidance to self-predicted features via GT-to-prediction annealing and visibility gating. With this interface fixed, different soccer behaviors can be learned by changing the local objective, geometry, and supervision that define the task.

DAVIS is designed to support a range of soccer skills through a shared learning framework: new skills are specified by task objects, commands, rewards, and curricula while retaining the depth-to-control interface. We evaluate goal-directed shooting and directional dribbling in detail as representative instances of one-shot striking and sustained ball control, using them to assess the feasibility of the framework across complementary contact regimes. Additional examples, including stopping the ball and obstacle-aware dribbling, are shown in Fig. 5; ball-loss recovery is demonstrated in the project-page video rather than as a panel in Fig. 5. Together, these instances provide evidence of the framework’s extensibility through task-specific training.

In summary, our contributions are as follows: (1) We introduce a **depth-only end-to-end RL framework** for humanoid soccer contact skills, where the deployed policy maps head-mounted depth, proprioceptive history, and low-dimensional task commands directly to 25-DoF joint targets with active-vision head control. (2) We develop a **visibility-aware auxiliary geometry mechanism** that learns task-relevant object geometry from the depth latent, and combines visibility masking with GT-to-prediction annealing to stabilize visual learning without adding runtime perception modules. (3) We instantiate multiple soccer skills within the same framework and establish a complete pipeline from IsaacLab training to E1 real-robot deployment, including depth alignment, policy export, and 25-DoF PD control. Real-robot evaluations and additional skill demonstrations support the framework’s practical feasibility and applicability across soccer tasks.

II. RELATED WORK

A. Robot Soccer Skill Learning

Robot soccer has long served as a challenging benchmark for robot control, requiring the integration of visual perception, real-time decision making, agile locomotion, and ball-oriented skills [15]–[17]. RoboCup-style soccer systems often relied on modular perception-control pipelines, where sensing, state estimation, and soccer control were separately engineered rather than learned end-to-end [18]. Recent learning-based methods have enabled legged robots to acquire soccer skills through various paradigms, including reinforcement learning and imitation learning [19]–[26]. Yet

most existing systems still depend on RGB-based object detection or multi-sensor perception pipelines [27]–[30].

In contrast, our work studies depth-driven humanoid soccer skill learning, where the robot learns to chase, dribble, and kick using only depth observations and proprioception.

B. Whole-Body Loco-Manipulation

Whole-body loco-manipulation on humanoid robots is inherently challenging, as the robot must simultaneously coordinate balance, locomotion, and task-directed manipulation within a high-dimensional and tightly coupled control space [31]–[33]. To make the problem tractable, prior systems often introduce decoupled or hierarchical interfaces, such as separate upper- and lower-body controllers or high-level task policies paired with low-level whole-body controllers [34]–[37]. Humanoid soccer further amplifies this difficulty: unlike manipulating static objects, the ball is a highly dynamic target, requiring the robot to approach from a suitable direction, maintain visibility, and generate precise foot–ball contacts for effective ball control.

To address these challenges, we propose a whole-body control framework, **DAVIS**, that jointly optimizes active visual tracking, approach, and foot–ball interaction for dynamic ball-centric loco-manipulation.

III. METHOD

A. Problem Formulation

We model humanoid soccer contact skills as a POMDP problem [38], denoting the process as $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma \rangle$. The simulator maintains a full state $\mathbf{s}_t \in \mathcal{S}$ containing robot, ball, goal, and contact-related physical variables; during training, privileged variables derived from this state, such as ball pose, goal pose, visibility, contact state, and reference motion, are used as critic inputs, reward-computation terms, and auxiliary-supervision labels. The deployed policy instead observes only a head-mounted depth image $\mathbf{D}_t$, a proprioceptive history $\mathbf{p}_{t-H:t}$ of length H , and an optional low-dimensional task variable $\mathbf{c}_t$; for instances without an external command, $\mathbf{c}_t$ is omitted. We write the deployed observation and action as $\mathbf{o}_t = (\mathbf{D}_t, \mathbf{p}_{t-H:t}, \mathbf{c}_t)$ and $\mathbf{a}_t = \pi_\theta(\mathbf{o}_t) \in \mathbb{R}^{25}$. The proprioceptive history is encoded inside the policy to infer latent body dynamics, while all explicit task geometry used by the actor must be produced from deployable inputs. The low-level controller converts the action into joint PD targets $\mathbf{q}_t^{\text{des}} = \mathbf{q}^0 + \mathbf{s} \odot \mathbf{a}_t$, where $\mathbf{q}^0$ is the nominal joint configuration and $\mathbf{s}$ is the per-joint action scale. The E1 humanoid consists of 23 body DoFs and 2 head DoFs, so we decompose the action as $\mathbf{a}_t = [\mathbf{a}_t^{\text{body}}, \mathbf{a}_t^{\text{head}}]$ with $\mathbf{a}_t^{\text{body}} \in \mathbb{R}^{23}$ and $\mathbf{a}_t^{\text{head}} \in \mathbb{R}^2$. Because the depth camera is mounted on the actuated head, the head action is part of the sensing loop: it directly changes the next depth observation. We optimize the asymmetric actor-critic policy with PPO [39] by maximizing the expected discounted return

$$\max_{\theta} \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^T \gamma^t (r_t^{\text{task}} + \lambda_{\text{amp}} r_t^{\text{amp}} - p_t^{\text{reg}}) \right], \quad (1)$$

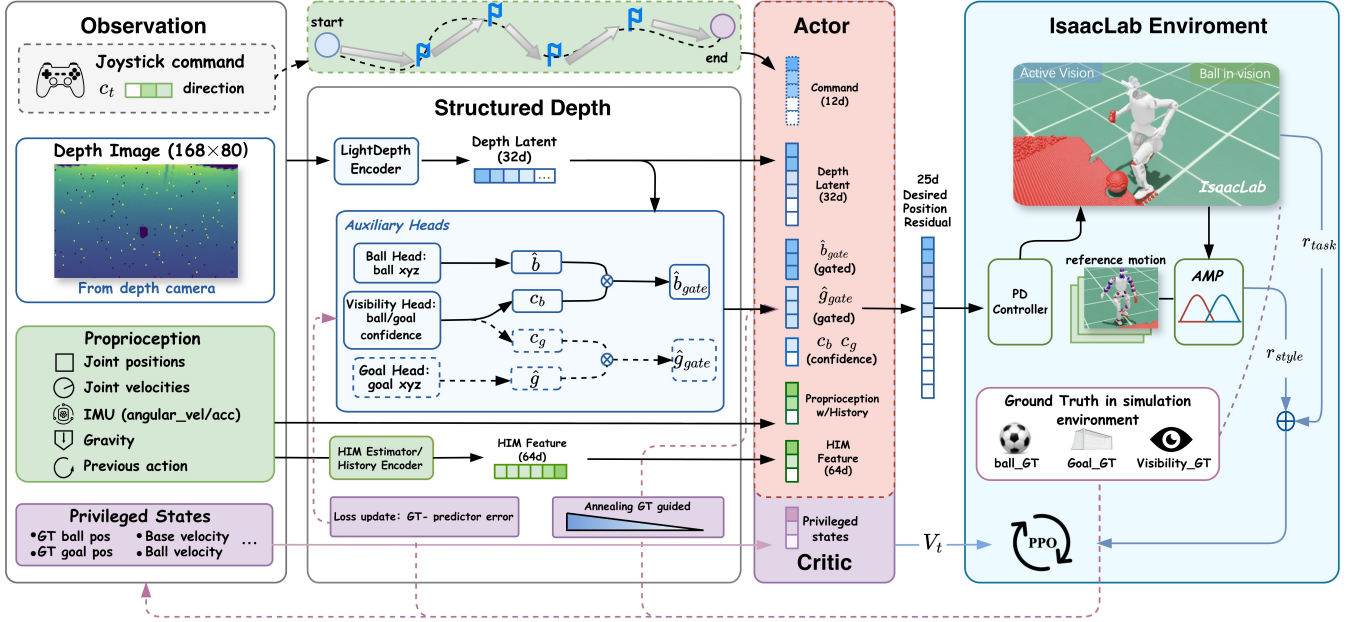


Fig. 2. **Overview of our DAVIS framework.** The observation stream contains the command c_t , a single 168×80 depth image, proprioceptive history, and privileged simulation states. The structured-depth module encodes the depth input into a 32-D latent, predicts ball/goal geometry and visibility confidence through **auxiliary heads**, forms confidence-gated geometry features, and combines them with a 64-D HIM/history feature. The actor receives the command, depth latent, gated geometry/confidence features, proprioceptive history, and HIM feature, and outputs a 25-D position residual for the PD controller. The auxiliary losses, GT-to-prediction annealing, PPO update, task reward, and style reward use simulation-only signals from IsaacLab rollouts, including ball, goal, and visibility ground truth. Dashed boxes in the structured-depth module indicate optional task-specific modules.

where r_t^{task} denotes task reward, r_t^{amp} encourages motion near a reference prior, and p_t^{reg} penalizes joint limits, action changes, unstable posture, collisions, and unsafe contact.

B. Depth-Only End-to-End Structured Soccer RL Framework

We build **DAVIS** around a deployment interface that remains deliberately narrow: the actor receives head-mounted depth, proprioceptive history, and an optional low-dimensional command, and outputs 25-DoF joint target residuals for the PD controller, as illustrated in Fig. 2. During training, privileged simulation states provide dense supervision for perception, value learning, and motion regularization; at deployment, the policy keeps only the depth/proprioception-to-control path.

To keep this deployment path compact, we introduce task structure only through training-time supervision and regularization. The depth latent feeds auxiliary geometry and visibility heads, whose predictions are confidence-gated before entering the actor. Simulation ground truth supplies geometry and visibility labels, but only as auxiliary supervision and as early-stage geometric guidance. A GT-to-prediction annealing schedule gradually replaces privileged features with predicted features, so the actor first learns how local geometry should affect whole-body motion and then operates with geometry inferred from depth. In parallel, PPO optimizes task return, the history estimator learns proprioceptive state inference, and AMP-style motion priors regularize full-body motion. These components make long-horizon soccer contact learning tractable while preserving a single deployable policy graph [40]–[42].

The next subsection specifies how task objects, variables, rewards, curricula, and optional contact conditioning in-

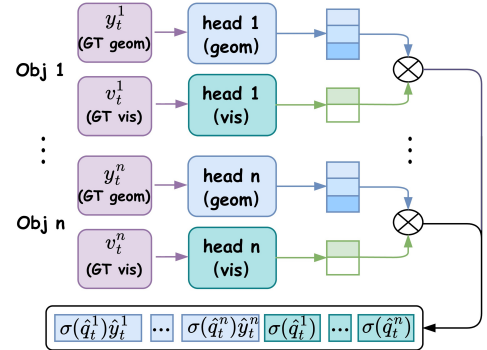


Fig. 3. **Scalable auxiliary geometry.** Each task object is assigned a geometry head and a visibility head supervised by privileged labels during training; their outputs are visibility-gated and concatenated into the actor feature, enabling the same mechanism to extend from one object to n objects.

stantiate individual skills while preserving this deployment interface.

C. Visibility-Aware Auxiliary Geometry

Learning task geometry from depth only through task returns is inefficient: task objects can occupy a small image region, while the gradient from PPO returns must pass through the actor and the depth encoder before shaping visual features. We therefore attach a visibility-aware auxiliary geometry module to the depth latent and supervise it with privileged simulation labels during training. As shown in Fig. 3, the module predicts object-level geometry and visibility inside the policy, so the actor receives direct object-level supervision without adding a runtime perception module.

Let $\mathbf{z}_t^D = f_D(\mathbf{D}_t) \in \mathbb{R}^{32}$ be the depth latent. For each task-relevant object $\text{obj}_i \in \mathcal{O}$, the auxiliary module predicts

its 3-D center and visibility confidence from the depth latent,

$$(\hat{\mathbf{y}}_t^i, \hat{q}_t^i) = h_{\text{aux}}^i(\mathbf{z}_t^D), \quad \hat{p}_t^i = \sigma(\hat{q}_t^i), \quad (2)$$

where $\mathbf{y}_t^i = [x_t^i, y_t^i, z_t^i]^\top$ denotes the object center in a right-handed head-yaw-aligned frame, with x along the head-yaw forward direction, y to the left, and z upward. Both geometry and visibility predictions depend only on the current 32-D depth latent; head states are provided separately to the actor through proprioception. The same construction extends to $\mathcal{O} = \{\text{obj}_i\}_{i=1}^n$ by assigning each object an indexed 3-D center prediction $\hat{\mathbf{y}}_t^i$, visibility prediction $\hat{q}_t^i$, ground-truth center $\mathbf{y}_t^i$, and visibility label v_t^i .

Visibility is learned jointly with geometry because a head-mounted camera can temporarily lose task-relevant objects. Geometry is supervised only when the corresponding object is visible, while visibility is trained with a binary objective,

$$\mathcal{L}_{\text{aux}} = \sum_{i=1}^n \lambda_i v_t^i \ell_i(\hat{\mathbf{y}}_t^i, \mathbf{y}_t^i) + \sum_{i=1}^n \lambda_{v_i} \text{BCE}(\hat{q}_t^i, v_t^i). \quad (3)$$

Visibility labels are computed using the full camera transform together with valid field-of-view and depth-range constraints; for multi-point objects, visibility is positive if any predefined structural point is valid, while the geometry target always remains the single 3-D object center.

To avoid forcing the actor to rely on unreliable predictions early in training, we gradually replace privileged geometry with predicted geometry. Let $\beta_k \in [0, 1]$ denote the prediction weight at policy update k . For object i ,

$$\mathbf{y}_{t,k}^{e,i} = (1 - \beta_k) \mathbf{y}_t^i + \beta_k \hat{\mathbf{y}}_t^i. \quad (4)$$

Using the correspondingly annealed visibility confidence $p_{t,k}^{e,i}$, we define the detached confidence gate

$$c_{t,k}^i = \text{clip}(\text{stopgrad}(p_{t,k}^{e,i}), 0.05, 1). \quad (5)$$

The actor-side auxiliary representation is then

$$\mathbf{u}_{t,k}^{\text{aux}} = \text{Concat}_{i=1}^n [c_{t,k}^i \mathbf{y}_{t,k}^{e,i}, c_{t,k}^i]. \quad (6)$$

As β_k increases from 0 to 1, the actor transitions from privileged geometric guidance to depth-based predictions. At deployment, all auxiliary features are predicted solely from depth.

D. Skill Instantiation

Once the deployment interface in Sec. III-B is fixed, a soccer skill is specified by a task definition rather than by switching runtime modules. This separation makes the interface reusable: new skills retain the depth-to-control path and are introduced by changing task objects, commands, objectives, and curricula. We denote such a definition by $\mathcal{T} = (\mathcal{O}, \mathbf{c}_t, \mathbf{y}_t, r_t, \mathcal{K})$, where $\mathcal{O}$ is the set of task objects, $\mathbf{c}_t$ is the optional command, $\mathbf{y}_t$ is the task geometry supervised through Sec. III-C, r_t defines the contact objective, and $\mathcal{K}$ is the curriculum. We use goal-directed shooting and directional dribbling as representative instances to verify the framework across complementary contact regimes; following details illustrate how the shared framework is instantiated for these examples.

1) *Goal-Directed Shooting*: Shooting is defined by the ball-goal geometry. Let $\mathbf{b}_t$ and $\mathbf{g}_t$ denote the ball position and goal center in the head-yaw-aligned frame, and let $\mathbf{d}_t^{bg} = \frac{\mathbf{g}_t - \mathbf{b}_t}{\|\mathbf{g}_t - \mathbf{b}_t\|_2 + \epsilon}$ be the normalized ball-to-goal direction. A behind-ball target can then be written as $\mathbf{x}_t^{\text{behind}} = \mathbf{b}_t - \rho \mathbf{d}_t^{bg}$, which turns chasing, alignment, and contact into one geometric target family. In this task, the object set is $\mathcal{O}_{\text{shoot}} = \{\text{ball}, \text{goal}\}$, the task geometry is $\mathbf{y}_t^{\text{shoot}} = (\mathbf{b}_t, \mathbf{g}_t)$, and the objective is to strike the ball so that its exit motion moves toward the goal. The reward is organized around run-up, contact, terminal success, visual maintenance, and safety regularization, while the curriculum gradually broadens the initial ball placement and approach difficulty. Active vision is maintained through the head DoFs: the head keeps the ball within view, and the body gradually follows the viewing direction.

2) *Directional Dribbling*: Dribbling is defined by a direction command and sustained ball regulation. Let $c_t \in \mathcal{C}$ be the commanded direction angle, where $\mathcal{C} = \{k\pi/6 : k = -6, \dots, 5\}$ is a 12-way ring uniformly spaced over 360° , and $\mathbf{d}_t^{\text{cmd}} = [\cos c_t, \sin c_t]$ the corresponding horizontal direction. The task object set is $\mathcal{O}_{\text{dribble}} = \{\text{ball}\}$, and the auxiliary geometry is the 3-D ball center $\mathbf{y}_t^{\text{ball}} = \mathbf{b}_t$. The commanded direction $\mathbf{d}_t^{\text{cmd}}$ and the robot-ball horizontal distance d_t are task variables used to define the dribbling objective rather than outputs of the auxiliary geometry head. Unlike shooting, the objective is not a single high-speed strike; the policy must keep the ball within a controllable band and generate progress along $\mathbf{d}_t^{\text{cmd}}$. We express the controllable-band condition by $\chi_t^{\text{band}} = \mathbb{I}(|d_t - d_0| < \Delta_d, v_{lo} < \|\mathbf{v}_t^b\| < v_{hi}, \mathbf{v}_t^b \cdot \mathbf{d}_t^{\text{cmd}} > 0)$. The reward covers commanded progress, re-approach, heading, visual maintenance, and out-of-band penalties. Continuous dribbling further uses contact-intent-conditioned AMP and phase masks so approach, touch, follow-through, and recovery emphasize different motion states.

E. Training and Deployment Pipeline

The deployment policy uses a single-frame head-mounted depth image, a five-step proprioceptive history, and task variables to output 25-DoF joint PD targets. Since this depth image is the main sim-to-real interface, we align simulation and deployment before it enters the actor. As shown in Fig. 4, a raw frame $\mathbf{D}_t$ is processed to $\tilde{\mathbf{D}}_t$. During training, the aligned frame is further perturbed to reduce dependence on clean depth and serve as domain randomization for sim-to-real transfer [43], [44]. Both simulated and real depth frames are cropped to 168×80 , clipped to 0.3–5.0 m, normalized, and zero-filled at invalid pixels. Policies are trained in Isaac Lab for 20,000 PPO iterations with AMP and a history-informed latent estimator (HIM) [45], then exported to ONNX. On the robot, the actor runs at 50 Hz, the low-level PD controller at 200 Hz, and the depth stream at 15 Hz; no external localization, object detector, state estimator, or runtime soccer planner is used.

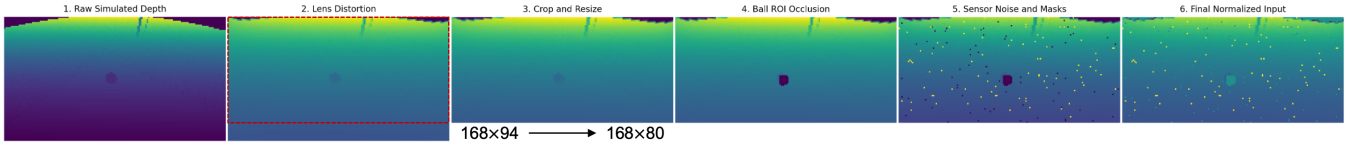


Fig. 4. Depth alignment transforms the raw depth frame into the aligned policy input through distortion correction, cropping from 168×94 to 168×80 , ball ROI occlusion, sensor noise, random masks and normalization, so the same single-frame depth interface can be used both in sim and real.

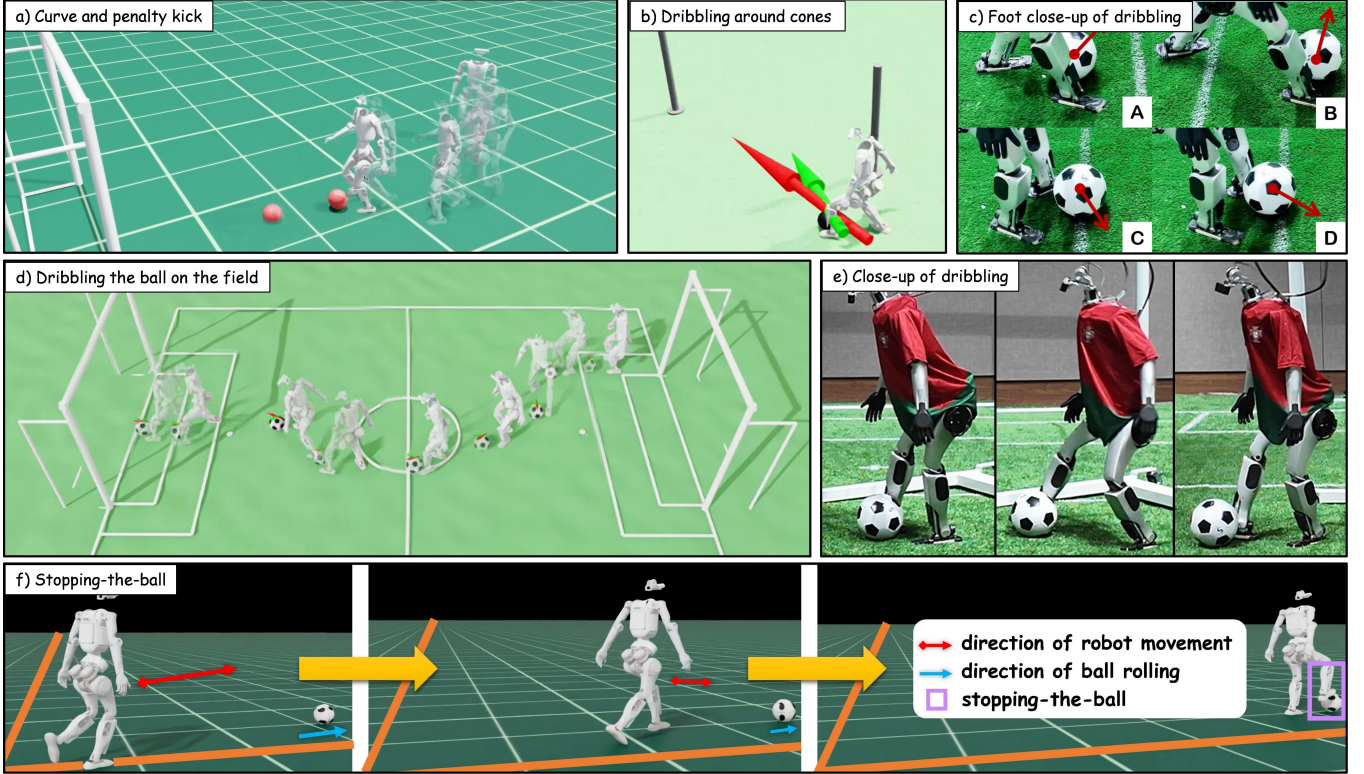


Fig. 5. **Qualitative examples of learned soccer skills.** In simulation: (a) curved and penalty kicks, (b) cone-avoidance dribbling, (d) dribbling across a soccer field, and (f) stopping a rolling ball, where red and blue arrows indicate robot and ball motion, respectively, and the purple box marks the stopped ball. On the real robot: (c) foot-ball contact during dribbling, with arrows indicating ball motion, and (e) a close-up dribbling sequence.

TABLE I: Quantitative results for goal-directed shooting and directional dribbling in simulation and on the real robot. Shooting entries and real-robot dribbling entries report success rate (SR) on a 0–1 scale; simulation dribbling reports SR, velocity error (E_{vel}), and ball-in-band ratio (R_{bib}) over 900 repeated-S slalom trials. Arrows indicate the preferred direction, and bold marks the best result within each fixed- N simulation-dribbling group. For shooting, a success requires the ball to cross the goal line in one kick. For dribbling, a success requires the robot to remain upright throughout execution and both the robot and ball to finish within 1 m of the target.

Goal-Directed Shooting						Directional Dribbling										
Setting	Metric	Easy	Med.	Hard	Total	Setting	Metric	N = 3			N = 5			N = 7		
								60°	90°	120°	60°	90°	120°	60°	90°	120°
Sim.	Penalty SR ↑	0.95	0.96	0.56	0.85	Sim.	SR ↑	0.71	0.77	0.65	0.66	0.72	0.55	0.68	0.64	0.43
	Free-kick SR ↑	1.00	0.88	0.71	0.85		E_{vel} ↓	0.69	0.74	0.91	0.74	0.73	0.80	0.71	0.81	0.97
	—	—	—	—	—		R_{bib} ↑	0.73	0.74	0.64	0.76	0.75	0.70	0.76	0.72	0.60
Real	Penalty SR ↑	0.68(26/38)	0.57(28/49)	0.55(33/60)	—	Real	SR ↑	Easy: 0.79(22/28)			Medium: 0.68(19/28)			Hard: 0.61(17/28)		

IV. EXPERIMENTS

We evaluate whether **DAVIS** supports both simulated and real-world soccer contact skills, as illustrated in Fig. 5. And we answer the following questions: **Q1**. Can a depth-only end-to-end policy perform soccer contact skills? **Q2**. Which training structures are necessary? **Q3**. Does the learned policy transfer to the real Noetix E1 robot?

A. Simulation Experiments

To answer Q1, we extensively evaluate **DAVIS** in IsaacLab simulator on two representative skill categories in soccer: shooting and dribbling.

Shooting and directional dribbling are trained as separate task-specific checkpoints that share the same depth-to-control interface; they are not a single multi-skill policy. At deployment, the shooting actor receives no operator command, while the dribbling actor receives only the intended direction from a joystick, quantized to the same 12-way ring defined in Sec. III-D.2.

Shooting: The robot is required to perform shooting with randomly initialized robot or ball positions. In the penalty kick task, the robot starts within an angle of $[-100^\circ, 100^\circ]$ behind the ball and distances of 2–3.5 m. In the free kick

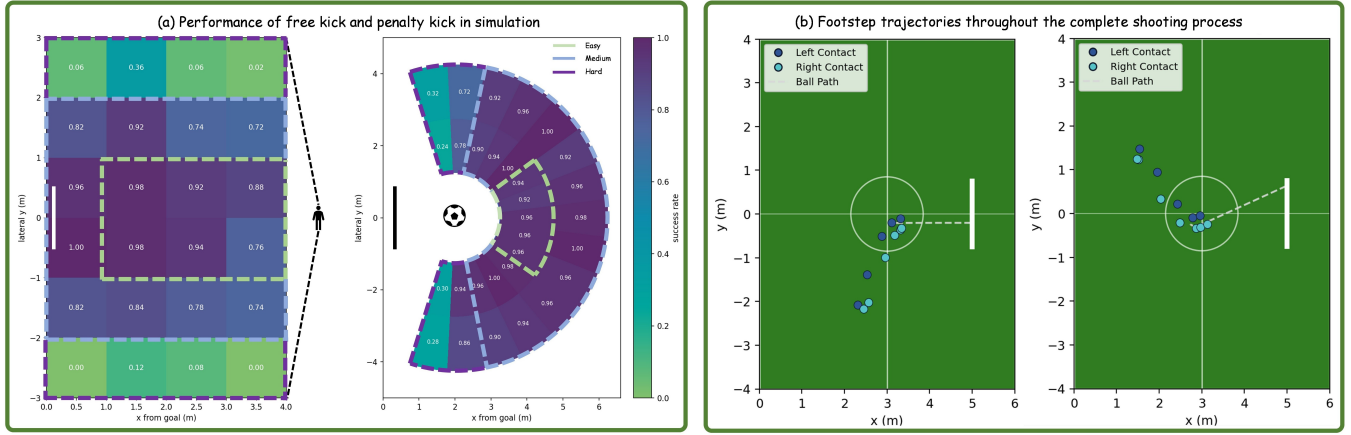


Fig. 6. (a) Free-kick and Penalty-kick performance in simulation. Each region uses over 50 trials. Darker colors indicate higher success rates, and dashed lines mark difficulty regions. (b) Footstep trajectories during shooting. The policy shows a smooth gait and consistent speed pattern.

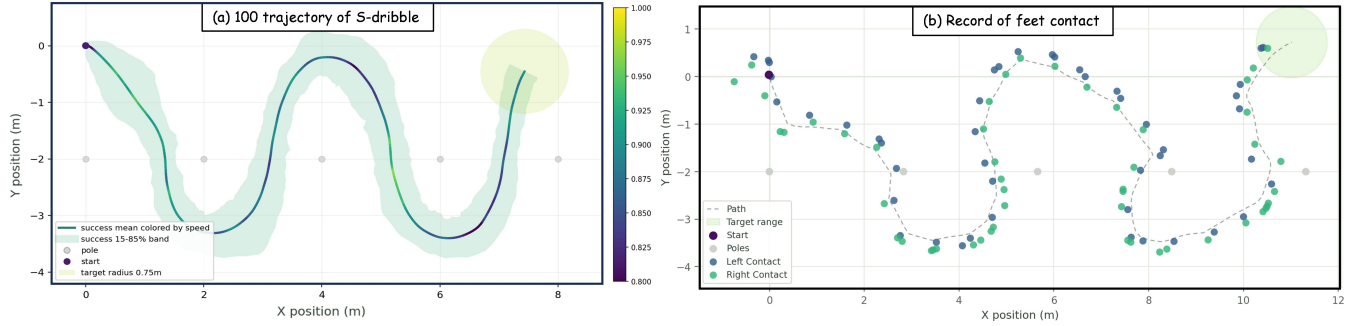


Fig. 7. Performance of repeated S-slam in simulation. (a) Statistical results of 100 S-slam trajectories. (b) Foot-contact record from a representative successful trajectory.

task, the robot is initialized 5 m in front of the goal, and is required to kick the ball within the region $x \in [0, 4]$ m and $y \in [-3, +3]$ m. The difficulty level categories are shown in Fig. 6. We record the success rate of the task. Experimental results in Table I demonstrate strong positional generalization capability. By analyzing the kinematic pattern during the shooting process in Fig. 6, we observe a smooth gait and consistent speed pattern.

Dribbling: The robot is required to complete a repeated-S slalom task. N is the number of poles, and α is the turning angle of instruction. A task is considered successful if the robot passes through all waypoints in the predefined order. To ensure consistency, we use a fixed instruction sequence for all experiments. The policy achieves an average success rate of 0.65 across 900 trials. Experimental results in Table I demonstrate that more than half of the task configurations obtain success rates above 0.65, showing robustness in long trajectory and extreme angles. The visualization in Fig. 7 shows the policy’s dynamic behavior.

Head-control comparison: We compare end-to-end learned head control with a heuristic that gazes at the predicted ball (Table II, last row). On static objects such as penalty kicks, the heuristic is comparable to learned active vision. On highly dynamic dribbling, the heuristic fails while the learned controller remains effective. The gap confirms that head motion must be learned jointly with locomotion and contact, rather than as a detached look-at-ball module, motivating active vision within the end-to-end policy.

Comparison with an external protocol: Dribble Mas-

ter [27] is the closest published system to our setting. Both use full-size humanoids with 2-DoF active heads and onboard sensing only; however, its policy consumes detector-derived ball coordinates, whereas DAVIS consumes a depth image. We evaluate DAVIS on its two core dribbling tasks with 50 trials per condition. At the 1 m tolerance, DAVIS achieves target-reaching and obstacle-avoidance success rates of 0.88 and 0.96, compared with the reported Dribble Master rates of 0.87 and 0.93.

B. Ablation Study

To verify the necessity of key structures and answer Q2, we conduct ablation studies on three tasks in simulation and record different metrics in Table II. All methods share the same training parameters and evaluation settings.

Ablation on curriculum. We remove the ramp-up and scaling curriculum. All reward terms are activated with full weights from the beginning. Without the progressive curriculum, the policy requires more training iterations to converge, and performance is significantly worse than baseline.

Ablation on auxiliary head. Our policy incorporates an object position prediction head. To validate effectiveness, we mask the predicted positions fed into the actor, while retaining the network architecture and input dimension. We disable the aux loss and stop backpropagation. As expected, without the explicit predictions, the model struggles to localize objects to make contact and success drops to zero. To further isolate each switch of the full auxiliary head, we disaggregate this into four finer variants: w/o anneal keeps

TABLE II: Simulation ablations and comparison. (a) Ablations. Penalty-kick and free-kick results use over 900 shooting trajectories, and repeated-S slalom uses over 100 dribbling trajectories. Metrics are success rate (SR), body-to-goal angular error (E_{ang}), successful ball-contact ratio ($r_{contact}$), velocity error (E_{vel}), and time in the control band (r_{bib}). The full method provides the best overall balance across tasks and metrics. (b) Comparison between our end-to-end learned head control (Active vision) and a heuristic head controller (Heuristic) that gazes at the predicted ball. The heuristic performs comparably to active vision in static-object tasks (e.g., penalty kicks), while active vision shows a clear advantage in highly dynamic dribbling scenarios.

Method	Penalty kick			Free kick			Repeated-S slalom		
	SR $\uparrow$	E_{ang} $\downarrow$	$r_{contact}$ $\uparrow$	SR $\uparrow$	E_{ang} $\downarrow$	$r_{contact}$ $\uparrow$	SR $\uparrow$	E_{vel} $\downarrow$	r_{bib} $\uparrow$
DAVIS	0.84	23.29	99.75	0.80	30.18	99.27	0.70	0.79	0.74
(a) Ablation									
DAVIS-w/o-curriculum	0.62	27.40	32.53	0.55	34.42	87.67	0.51	0.77	0.69
DAVIS-auxiliary head-w/o-anneal	0.32	54.79	84.87	0.22	76.20	87.92	0.31	0.83	0.66
DAVIS-auxiliary head-w/o-geometry	0.20	53.53	87.33	0.15	69.47	78.13	0.34	0.91	0.54
DAVIS-auxiliary head-w/o-supervision	0.13	81.87	93.33	0.14	93.66	91.67	0.30	0.95	0.48
DAVIS-auxiliary head-w/o-visibility gate	0.34	51.97	91.33	0.29	68.44	92.08	0.27	1.06	0.64
DAVIS-w/o-auxiliary head	0	–	0	0	–	0	0	1.45	0.05
DAVIS-w/o-active vision	0	–	0	0.36	97.81	1.71	0	1.25	0.03
(b) Comparison									
DAVIS-Head controller	0.88	26.66	93.67	0.76	37.96	88.13	0.0	0.99	0.13

$\beta_k = 1$ throughout (always predicted auxiliary features; no GT $\rightarrow$ pred schedule); w/o geometry zeros the actor’s auxiliary input while keeping auxiliary losses; w/o supervision drops the goal/ball/visibility losses but still feeds predicted features; w/o vis. gate skips visibility $\times$ geometry masking.

Ablation on active vision. Active vision is another key design. Since its components all act through the same two head-joint control dimensions and cannot be ablated independently, We fix the 2-DoF head joints, replace the original head-based rewards with body alignment rewards, and adjust the curriculum. After fixing the head joints, the ball frequently moves out of the FOV or enters blind spots, causing collapse. Another notable observation is that the robot is unable to locate the gate. The performance drops nearly to zero, with a high likelihood of falling.

C. Real-World Deployment

To answer Q3, we deploy our policies on the Noetix E1 humanoid robot equipped with a ZED2i depth camera mounted on a 2-DoF head. For the shooting experiments, we evaluate 12 predefined positions in one penalty-kick protocol; the Easy, Medium, and Hard settings use progressively more distant and more oblique positions, with 38, 49, and 60 repeated trials, respectively. For dribbling experiments, we design three tasks including reaching target position, obstacle traversal, and slalom dribbling. To better reflect real-world soccer scenarios, all experiments are conducted on a grass surface. Safety protection uses a tether or staff member; emergency-stop and projected-gravity fall triggers terminate unsafe rollouts, which are restarted from their initial conditions. If the ball is lost without an active recovery behavior, the contact policy exits to the default walk-run policy. Fig. 1 and Fig. 5 show that our policy can perform dynamic soccer tasks with precise actions. Quantitative results in Table I demonstrate that our policy maintains a high success rate in real deployment. Shooting records 26/38, 28/49, and 33/60 successes for the Easy, Medium, and Hard settings, respectively; a success requires the ball to cross the goal line in one kick. Dribbling records 22/28, 19/28, and 17/28

successes for target reaching, obstacle traversal, and slalom, respectively; a successful dribbling trial requires the robot to remain upright throughout execution and both the robot and ball to finish within 1 m of the target. All dribbling trials were conducted by the same operator using a joystick controller.

V. CONCLUSION

We presented a depth-only end-to-end active-vision framework **DAVIS** for humanoid soccer skills. The main idea is simple: keep the deployment interface minimal, but place enough structure inside training so that the policy can learn soccer-specific geometry, visibility, and whole-body contact behavior from depth and proprioception alone.

Within the shared framework, goal-directed shooting and directional dribbling serve as representative instances of one-shot striking and sustained ball control, retaining the observation-to-control interface while changing task-specific objects, commands, rewards, curricula, and motion-prior conditioning. Across simulation, ablations, and the complete Noetix E1 deployment pipeline, real-robot execution confirms that this depth-only interface transfers across contact regimes; together with additional skill demonstrations, these results support DAVIS as an extensible framework for humanoid soccer skills through task-specific training.

VI. LIMITATIONS AND FUTURE WORK

Two limitations remain. The first is perceptual: depth-only sensing keeps the interface compact but ties the policy to a single head-mounted camera, inheriting its occlusions, dropouts, and imperfectly simulated depth—a residual sim-to-real gap that limits real-world accuracy. The second sits upstream: the task-definition interface unifies shooting and dribbling, yet each skill needs hand-tuned rewards, curricula, and contact conditioning, leaving the pipeline a recipe, not an automatic one.

Closing the perceptual gap is largely engineering—sensor-noise modeling and real-world fine-tuning. Automating skill construction is the more open problem, where language-conditioned specification and automatic curricula could generate new behaviors on the same interface.

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